



PREPARING HISTORICAL ASSET NAMING DATA FOR A PUBLIC GIS MAP

From asset folders and historical reports to structured map records

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Why the asset naming map matters

Make the history behind City asset names easier to discover.

For people

Find a named asset, see its location, and learn the story connected to its name.

For City records

Connect map information to reports, approvals, and images in the asset folder.



A century of records in many formats



Historical files reviewed

1890 to 2026

Communities, parks, bridges, roadways, CTrain stations, recreation facilities, plazas, operational facilities, and other named places.

More than a century of naming history

The Excel workbook became the project backbone

I created one structured record for each asset and used it to connect the map entry to the correct folder and documents.

Asset Name	Category	Source	Approval Date	Community	Quadrant	Renaming	Theme	Report	Approval	Outstanding	Map Status
Cowboys Park	Park	Council	06/18/2024	Downtown	SW	Yes	Sponsorship	Report	Approval		Loaded
Peace Bridge	Bridge	Council	07/27/2009	Sunnyside	NW	No	N/A	Report	Approval		Loaded
City Centre Parkade	Parkade	Council	07/03/2000	Beltline	SW	No	Functional	Report	Approval		Loaded

The workbook is the shared input and control file for the automation scripts. It also documents field definitions, naming conventions, future phases, outstanding items, and loading status.

Building the Excel dataset



1. Crawl historical folders

Python scanned year and asset folders, extracted recognizable information, detected dates, and identified document types.

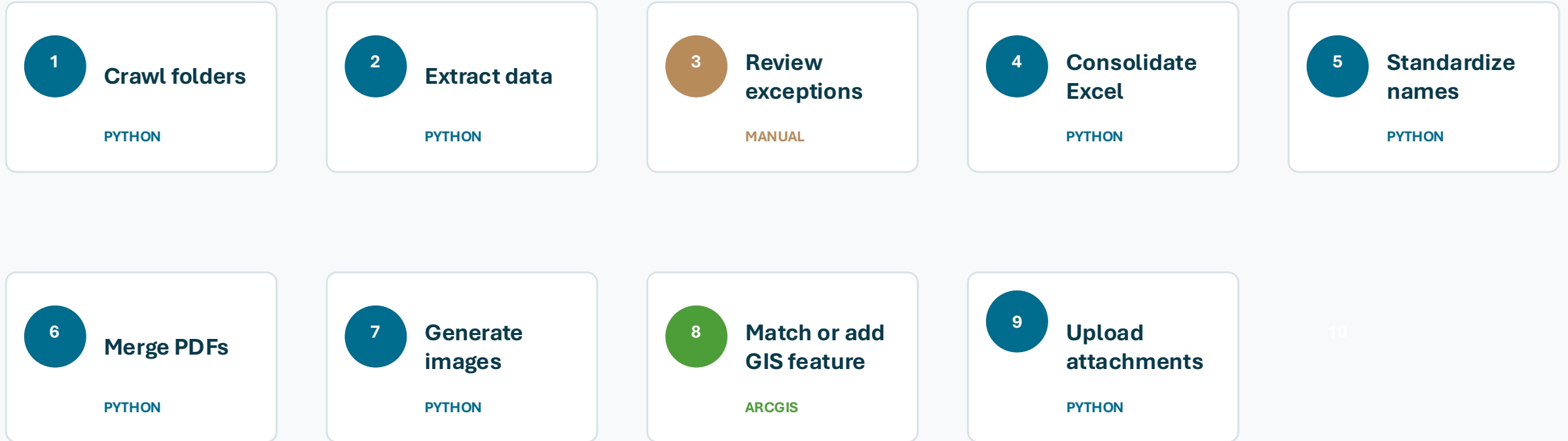
2. Consolidate results

Yearly extraction results were combined into the main Excel workbook used across the project.

3. Review exceptions manually

Handwritten, incomplete, missing, or incorrectly interpreted information was checked and corrected.

Automation across the workflow



Automation reduced repetitive work. Manual review remained essential for historical exceptions.

Building a reusable Python automation workflow

Data extraction and consolidation

Crawled historical folders; extracted names, dates, categories, and document types; created yearly results; combined results into the main workbook.

File and document preparation

Organized selected and extra files; matched reports and attachments; merged PDFs; renamed reports and approvals using Excel conventions.

Image preparation

Generated standardized asset-name title images from Excel; distributed the images into matching asset folders; supported individual corrections.

GIS upload and reporting

Matched or created GIS features; updated attributes; uploaded images before PDFs; tracked processed, unmatched, failed, and skipped outcomes.

Standardizing and preparing PDFs



Different folder structures

Some assets had one file; others had a main report, multiple attachments, and a separate approval document.

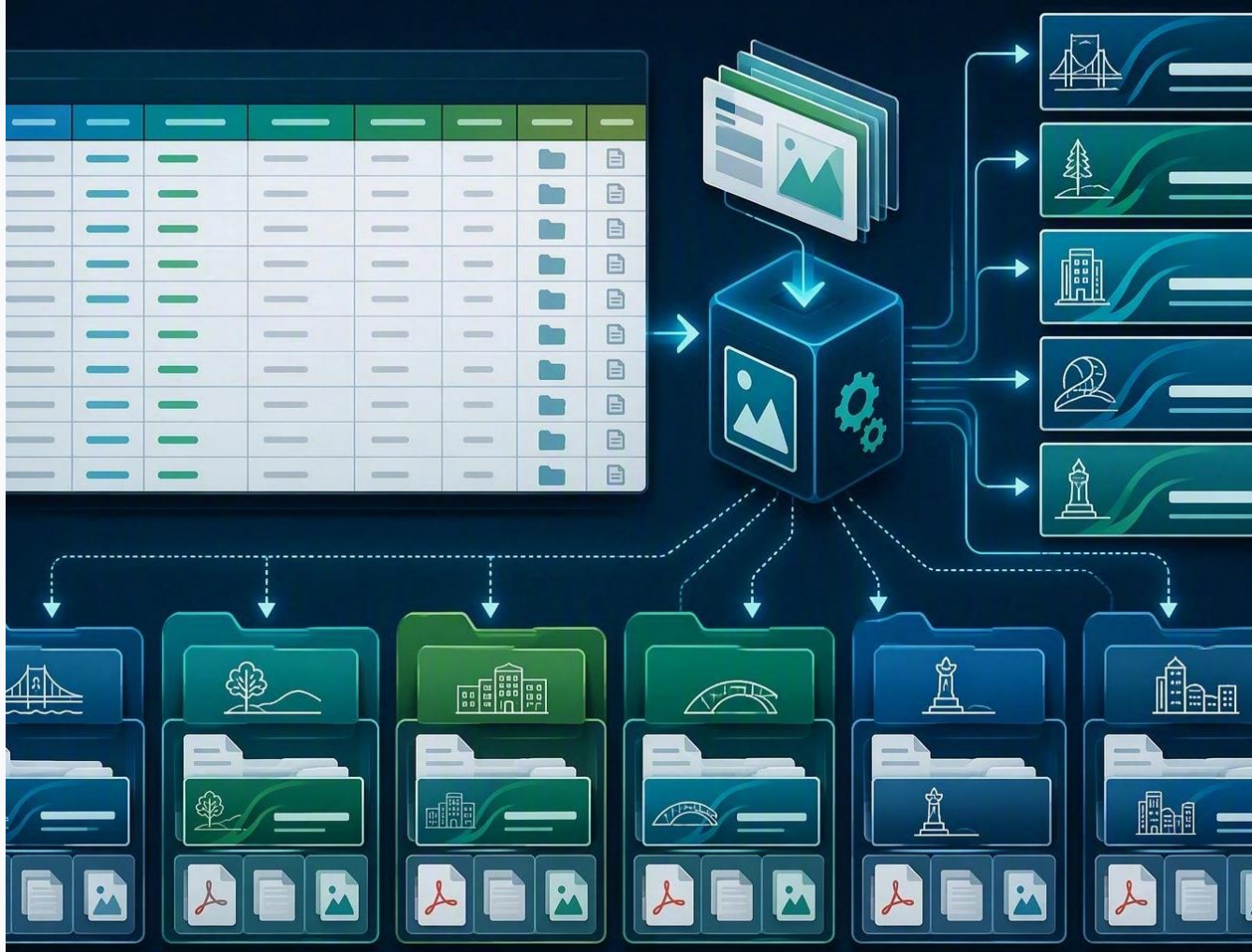
Two upload document types

Naming/Sponsorship Report and Naming/Sponsorship Approval Document.

Python-supported preparation

Scripts identified main reports and numbered attachments, merged related files, renamed PDFs from Excel, and flagged folders needing review.

Generating and distributing asset-name images



Each map record also needs a title image.

- 1 Read asset names from Excel
- 2 Generate standardized title images
- 3 Match and distribute to asset folders
- 4 Upload images before PDFs

Uploading prepared assets to the map



**ASSET ALREADY
ON THE MAP?**

Yes: update

Match by asset name and approval date, update attributes, and upload the prepared image and PDFs.

No: create

Search park or community source layers, derive a point from available geometry, and create the feature.

Safe testing: authentication check > one-row test > dry run > full update

Progress to date

Source records

1890 to 2026

Map upload completed so far

1984 to 2026

The workbook and CSV logs track loaded, verified, not found, failed, skipped, and outstanding records.



Next: continue with records before 2000

1,713
asset records prepared

7,494
supporting PDFs standardized

Challenges in the older records

Poor or changing formats

Older years do not consistently use the same structure, and current fields were not always available.

Handwritten documents

Some records are difficult to read and cannot be extracted reliably by a script.

Missing information

Details may be absent because the format was not followed or had not yet been introduced.

Unclear file relationships

Folders may contain several reports, approvals, or attachments without a consistent pattern.

Historical variation limits full automation. Python improves consistency, while human review protects accuracy.

What I learned

01 **Automation reduces repetitive work** Python supported crawling, consolidation, organization, PDF merging, renaming, image generation, GIS updating, attachment upload, and logs.

02 **Manual verification protects accuracy** Historical records need interpretation when text is unclear or important information is missing.

03 **Standards support future maintenance** Consistent Excel fields, folder names, and PDF names make ongoing work easier.

04 **Progress must remain traceable** Status and outstanding-item columns separate completed records from questions.

Asset Naming: public-facing example



Explore the map

The application connects mapped assets with structured naming information and supporting documentation.

[OPEN LIVE MAP](#)

FROM HISTORICAL ASSET NAMING DOCUMENTS TO A PUBLIC GIS MAP

My contribution was to make the records consistent, connect each asset to its documents, and prepare the information for public access on the map.

Thank you

Questions and discussion

